

Kuiyi (Cory) Gao

Chapel Hill, NC | kuiyigao@unc.edu | kuiyigao.github.io | [GitHub](#)

Mandarin (native) | English — TOEFL iBT 110 | Cantonese (intermediate)

Research Interests

AI safety and auditable deployment; multimodal safety evaluation; agent robustness and runtime monitoring; human oversight and value alignment.

Education

University of North Carolina at Chapel Hill

Jan 2026 – present

Undergraduate; intended B.A. in Computer Science

Expected May 2027

- GPA: 3.68/4.00. Exchange student in Spring 2026; transferred in August 2026.
- Coursework: COMP 550 Algorithms and Analysis, COMP 562 Machine Learning, COMP 586 Natural Language Processing, COMP 380 Technology, Ethics & Culture.

The Chinese University of Hong Kong

Sep 2023 – May 2026

Undergraduate study in Computer Science; transferred to UNC

- Faculty of Engineering ELITE Stream. Coursework: ELEG 5491 Introduction to Deep Learning (graduate level), ESTR 3108 Artificial Intelligence, Data Structures, Formal Languages & Automata Theory, Computer Organization & Design, Probability & Statistics, Linear Algebra.

Publications and Manuscripts

[1] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 (Main Conference). [\[paper\]](#) [\[code\]](#)

[2] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [\[paper\]](#) [\[code\]](#)

[3] Co-first-authored manuscript on *GUI-agent robustness under interface noise*. Under review, 2026.

[4] First-authored manuscript on *RuntimeTickets for third-party agent skills*. Under review, ICLR 2027.

Research Experience

GUI-Agent Robustness under Interface Noise [3]

Hong Kong & remote

CUHK; advisor: Prof. Michael R. Lyu

Sep 2025 – present

- **Co-first author.** Co-developed a benchmark for web and Android tasks with interface changes such as overlays, moving content, and altered layouts; compared task completion, unsafe actions, and action sequences.
- Designed metrics for success within a retry limit, unsafe actions, and changes in action sequences; built the automated judging pipeline and ran experiments on paired original and altered interfaces.

RuntimeTickets and Agent Skill Firewall [4] [\[code\]](#)

Abu Dhabi & Beijing

MBZUAI UGRIP and continuing research; advisor: Dr. Steve Liu

Jun 2026 – present

- Built the runtime monitor in a four-student UGRIP team. It compared declared skill behavior with observed activity and produced PASS, DEFER, or BLOCK decisions with supporting logs.
- **First author.** Leading method development, experiments, and writing for RuntimeTickets. Its records link approved permissions, observed activity, and safety decisions for independent checking.

Research Experience (continued)

Chain-of-Jailbreak: Image Generation Safety [2]

Tencent AI Lab & CUHK

Hong Kong

Jun 2024 – Feb 2025

- **Second author.** Designed the method and a prompting-based defense; contributed benchmark data and ran most attack and defense experiments on multi-turn image editing.
- Built model-output evaluation pipelines, prepared paper figures, and contributed to the manuscript and reviewer responses.

MMSafeAware: Multimodal Safety Awareness [1]

Tencent AI Lab

Hong Kong & Shenzhen

Dec 2023 – Oct 2024

- **Third author.** Contributed evaluation design and benchmark construction to a study of 1,500 image-prompt pairs across 29 safety scenarios, covering unsafe and benign inputs.
- Ran vision-centric reasoning fine-tuning experiments; evaluated misses on unsafe inputs separately from false alarms on benign inputs, and contributed to submission and rebuttal. The overall study covered nine models and three mitigation approaches.

Selected Projects

MemSlot: Saliency for Long-Document Compression [code]

2026

UNC COMP 586, Natural Language Processing

- Implemented memory-slot attention over frozen RoBERTa features in PyTorch, using reconstruction and slot-diversity losses; merged scores from overlapping token windows into word-level saliency.
- Built saliency-guided rendering of CUAD contracts and an OCR-to-text QA pipeline with a separate reader, plus text-pruning and summarization baselines. End-to-end evaluation is not yet complete.

Poetry and Image Variation

2025

CUHK ENGE3940, AI and Language

- Analyzed saved CLIP-based image-variation scores for ten Imagist and Romantic poetry excerpts, with 20 images per excerpt from each of FLUX.1-dev and Stable Diffusion.
- Used Python to plot score distributions, compare poem groups, and examine the relationship between prompt length and image variation.

Text-KEN: Kernel Comparisons for Sentence Embeddings

May – Aug 2025

CUHK summer research; advisor: Prof. Farzan Farnia

- Built a Python prototype to compare two sets of MiniLM sentence embeddings using RBF kernels and spectral entropy; explored centering and covariance whitening.

Technical Skills

Programming and data Python, NumPy, pandas, SciPy; Java and C (coursework).

ML and NLP PyTorch, Hugging Face Transformers, sentence-transformers; attention modules, frozen encoders, and fine-tuning experiments.

Evaluation Benchmark construction; task completion and action-sequence metrics; embedding comparisons; span-QA exact match/F1.

Visualization Matplotlib, seaborn, Pillow; statistical plots, document rendering, and image grids.

Research tools Docker, Playwright, Git, HTTP/JSON APIs; experiment configuration and trace logging.

Honors and Awards

Professor Joseph Lau Scholarship of Innovation; CUHK College Entrance Scholarship; CUHK exchange scholarship (UNC, Spring 2026).